

# Lab 7

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2022-11-14

**Deadline: 2022-11-20 22:30:00 Beijing Time**

# How to submit?

Email: [YJ40202@163.com](mailto:YJ40202@163.com)

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Attachment: Your java source files.

Note: Don't use package.

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# Exercise07\_11.java

The standard deviation of n numbers is defined as

$$\bar{x} = \frac{1}{n} \sum_{i=1}^n x_i \quad \sigma = \sqrt{\frac{1}{n-1} \sum_{i=1}^n (x_i - \bar{x})^2}$$

Your program should contain the following methods:

```
public static double mean(double[] x) // Compute the mean of an array of double values
public static double deviation(double[] x) //Compute the deviation
```

Write a test program that prompts the user to enter 10 numbers and displays the mean and standard deviation, as presented in the following sample run

```
Enter 10 numbers: 1 2 3 4.5 5.6 6 7 8 9 10 [Enter]
The mean is 5.61
The standard deviation is 2.997943739743404
```

## Exercise07\_15.java

(Eliminate duplicates) Write a method that returns a new array by eliminating the duplicate values in the array using the following method header:

```
public static int[] eliminateDuplicates(int[] list)
```

Write a test program that reads in 10 integers, invokes the method, and displays the distinct numbers separated by exactly one space. Here is a sample run of the program:

```
Enter ten numbers: 1 1 22 22 3 3 44 44 5 5 [Enter]  
The distinct numbers are: 1 22 3 44 5
```

# Exercise07\_19.java

(Sorted?) Write the following method that returns true if the list is already sorted in ascending order.

```
public static boolean isSorted(int[] list)
```

Write a test program that prompts the user to enter a list and displays whether the list is sorted or not. Here is a sample run. Note that the first number in the input indicates the number of the elements in the list. This number is not part of the list.

```
Enter a list: 2 1 2
The list is already sorted
```

```
Enter a list: 4 1 3 2 5
The list is not sorted
```

## Exercise07\_33.java

(Chinese Zodiac) The Chinese Zodiac is based on a 12-year cycle, with each year represented by an animal: rat, ox, tiger, rabbit, dragon, snake, horse, sheep, monkey, rooster, dog and pig. 1900 is the year of the rat.

Write a program that prompts the user to enter a year and displays the animal for the year. Here are some sample runs:

```
Enter a year: 2020 [Enter]  
rat
```

```
Enter a year: 1866 [Enter]  
tiger
```

# Exercise08\_01.java

(Sum elements row by row) Write a method that returns the sum of all the elements in a specified row in a matrix using the following header

```
public static double sumRow(double[][] m, int rowIndex)
```

Write a test program that reads a 3-by-4 matrix and displays the sum of each row. Here is a sample run:

```
Enter a 3 by 4 matrix row by row:
1.5 2 3 4
5.5 6 7 8
9.5 1 3 1
Sum of the elements at row 0 is 10.5
Sum of the elements at row 1 is 26.5
Sum of the elements at row 2 is 14.5
```



## Exercise08\_05.java

(Algebra: add two matrices) Write a method to add two matrices. The header of the method is as follows:

```
public static double[][] addMatrix(double[][] a, double[][] b)
```

Write a test program that prompts the user to enter two 2 \* 2 matrices and displays their sum. Here is a sample run:

```
Enter matrix1: 1 2 2.2 3
Enter matrix2: 10 1.5 2 7
The addition of the matrices is
1.0 2.0      10.0 1.5      11.0 3.5
2.2 3.0  +   2.0 7.0  =   4.2 10.0
```